

Siao Xu

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Research Fields

Panel Data, Financial Econometrics, Empirical Bayes, Machine Learning, Time Series and Network Science.

Academic Employment

ShanghaiTech University, China
Research Assistant

Sep 2019 – Jun 2021

Education

Universität Mannheim, Germany
Ph.D. in Economics

Sep 2021 – 2027(Expected)

Hong Kong University of Science and Technology, Hong Kong
M.Sc. in Financial Mathematics

Sep 2016 – Jun 2018

London School of Economics, UK
B.Sc. in Mathematics and Economics

Sep 2012 – Jun 2015

Job Market Paper

[Job Market Paper] **Machine-Learning-Assisted Empirical Bayes Estimation for Regression on grouped Network Data**

Abstract: We propose a two-way effects model for matched data represented by a bipartite network with latent group structure. A directed acyclic graph (DAG) encodes the assumed relationships among the observed outcomes, network links, latent effects, group memberships, and prior parameters, thereby clarifying the data-generating process and organizing the empirical Bayes (EB) estimation procedure. Unobserved heterogeneity is modeled using group-specific prior distributions, allowing individual effects to vary both across and within groups. Although the two-way effects can be estimated by least-squares estimator, LS may be imprecise when the underlying bipartite network is sparse and connected across latent groups. We therefore develop a machine-learning-assisted EB estimator that first learns the latent group structure from the network and then exploits this structure to shrink the estimated effects toward their corresponding group-specific distributions. We establish the theoretical properties of the proposed estimator, and Monte Carlo simulations show that it substantially reduces estimation error relative to OLS under the data-generating processes considered. Finally, in an empirical application to Amazon user-product ratings, we use the proposed method to estimate and characterize heterogeneity in the user- and product-specific components of ratings.

Working Paper

[1] **Mixed Membership Estimation in Bayesian Network Autoregression**
with Endong Wang, *R&R* at The Econometrics Journal

[2] **Mixed Membership Estimation in Partial Correlation Network**
with Endong Wang, under review at Journal of Applied Econometrics

[3] **The Horizon Structure of Contagion: High-Dimensional Measurement and Inference**
with Abderrahim Taamouti and Endong Wang

- [4] A New Empirical Bayes Estimation for Network Peer Effect Model
- [5] Overlapping Community Detection in Mixed Membership Vector Autoregression

Work in Progress

- [1] Bi-group Detection in Large Matrix-Variate Factor Model
with Endong Wang
- [2] Mixture Prior on Random Coefficient Model

References

Prof. Carsten Trenkler Department of Economics Universität Mannheim Email: trenkler@uni-mannheim.de	Prof. Mengshan Xu Department of Economics Universität Mannheim Email: mengshan.xu@uni-mannheim.de
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Prof. Endong Wang
Department of Economics
Universität Mannheim
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Conference, Workshop and Seminar

[2026] 2026 CSW Asia Meeting (Main) in Abu Dhabi, ENTER Jamboree in Madrid, Mannheim Econometrics Seminar, Time Series AI in Macroeconomics and Finance in Bristol, Bristol Econometric Study Group Conference, 31st International Panel Data Conference in Exeter, 2026 European Summer Meeting (ESEM) in Dublin, UCL PHD seminar.

[2025] (EC^2) Conference in Lugano, 45th International Symposium on Forecasting, 9th SETA (Macau), Mannheim Econometrics Seminar, HKMetrics Workshop in Mannheim, 2025 Harbin International Conference on Econometrics and Statistics, GESS Research Day in Mannheim.

[2024] Workshop in Asian Summer School of Econometrics, Mannheim Econometrics Seminar.

Teaching

[Fall 2026] CC502 Applied Econometrics (Master Level), Universität Mannheim

[Fall 2025] BE510 Business Economics 1 (Master Level), E700 Mathematics for Economists (Doctor Level), Universität Mannheim

[Fall 2023] BE510 Business Economics 1 (Master Level), Universität Mannheim

[Spring 2023] Macroeconomics A (Bachelor Level), Universität Mannheim

[Fall 2022] CC502 Applied Econometrics (Master Level), Universität Mannheim

Technologies

Programming: Python, R, Matlab, Stata, C++, C, Java

Language: Mandarin (native), English (fluent)

Nationality: Chinese